

Section A

1	(a) (i) $F = GMm / R^2$	B1	[1]
	(ii) $F = mR\omega^2$	B1	[1]
	(iii) reaction force = $GMm / R^2 - mR\omega^2$ (allow e.c.f.)	B1	[1]
	(b) (i) <i>either</i> value of R in expression $R\omega^2$ varies or $mR\omega^2$ no longer parallel to GMm / R^2 / normal to surface becomes smaller as object approaches a pole / is zero at pole	B1 B1	[2]
	(ii) 1. acceleration = $6.4 \times 10^6 \times (2\pi / \{8.6 \times 10^4\})^2$ = 0.034 m s^{-2}	C1 A1	[2]
	2. acceleration = 0	A1	[1]
	(c) e.g. 'radius' of planet <u>varies</u> density of planet <u>not constant</u> planet spinning nearby planets / stars (any sensible comments, 1 mark each, maximum 2)	B2	[2]
2	(a) (Thermal) energy / heat required to convert unit mass of solid to liquid	M1	